Ira A. Fulton Schools of Engineering
Arizona State University

Interdisciplinary Capstone Project Proposal Request Form
Fall 2025/Spring 2026

Partner Contact		Project Contact (If Different)	
Company	General Dynamics Mission Systems	Company	General Dynamics Mission Systems
Name	Glen Abousleman	Name	Glen Uehara
Email	Glen.Abousleman@gd-ms.com	Email	Glen.Uehara@gd-ms.com
Phone	480-441-2193	Phone	480-441-5064

Project Title

Enhancing Security on Embedded Systems Using TinyML/EdgeAI

Project Summary

Project description: The objective of this capstone project is to develop and deploy a TinyML/EdgeAI model that can detect and respond to security threats on embedded systems using data from sensors, thereby enhancing security.

The components of this project are as follows:

- Data Collection:
  - Sensor Deployment: Install sensors on the embedded system to monitor various parameters such as CPU usage, memory usage, network traffic, and system logs
  - Data Logging: Collect and log sensor data over a period to capture both normal and abnormal system behaviors, including known security threats
- Data Preprocessing:
  - Data Transfer: Transfer the collected data from the embedded system to a large server for preprocessing and analysis
  - Feature Engineering: Extract and preprocess relevant features from the raw sensor data, focusing on indicators of potential security threats
- Model Development:
  - Model Selection: Choose a suitable machine learning model (e.g., anomaly detection, classification) based on the nature of the security threats
  - Training on Server: Use the large server to train the selected model on the preprocessed data, taking advantage of its computational power to handle large datasets and perform extensive hyperparameter tuning
  - Optimization for embedded platform: Optimize the trained model to ensure it is lightweight and can run efficiently on the embedded

system. This may involve model pruning, quantization, and other techniques

- Model Deployment:
  - ML/AI Framework: Convert the optimized model into a format compatible with small ML/AI frameworks such as TensorFlow Lite for Microcontrollers
  - Deployment on Embedded System: Deploy the model onto the embedded system where the sensors are running
  - Integration: Integrate the model with the embedded system's security infrastructure to enable real-time data analysis and threat detection
- Real-time Analysis:
  - Sensor Data Processing: Continuously feed sensor data into the deployed model for real-time analysis
  - Threat Detection and Response: Implement mechanisms to respond to detected threats, such as alerting administrators or taking automated protective actions
- Testing and Validation:
  - Scenario Testing: Test the deployed model using various scenarios to validate its accuracy and performance in detecting security threats
  - Model Tuning: Fine-tune the model based on testing results to improve detection capabilities and reduce false positives/negatives
- Performance Evaluation:
  - Resource Utilization: Evaluate the impact of the model on the embedded system's resource utilization (CPU, memory, power consumption)
  - Efficiency: Ensure the model operates efficiently within the constraints of the embedded system
- Documentation and Reporting:
  - Process Documentation: Document the entire project process, including data collection, model development, and deployment steps
  - Final Report: Prepare a final report detailing the findings, performance metrics, and potential areas for further improvement

Student Learning Experience: Students will gain first-hand experience with the following concepts:

- TinyML/edgeAI development and understanding: A trained tinyML/edgeAI model capable of real-time security threat detection on an embedded system using sensor data
- Cyber security, zero trust: Enhanced security for the embedded system through proactive and efficient threat detection and response
- Project/Program life cycle: Project responsibility, accountability and delivery

Deliverables: Project deliverables are phased through the two-semester sequence and include

- Comprehensive documentation of the project process and results
- Any source code or model
- If progress is being made, GDMS will work with supplier to provide next generation processor to validate the model and algorithm

Required background: Background information required to successfully execute the project includes:

- Python
- Machine Learning
- Embedded System
- Project Management/Organization (a plus)
- Using Agile development (Scrum, Kanban, CI/CD, pipelines, Sprints) – a plus

Supplies/Equipment Required:

- Computer for development and documentation
- GitHub side (public site is ok, make it private so only those on program can access)
- Access to ASU supercomputer (e.g. Sol)

Engineering Disciplines Required	None	Low	Med	High	Engineering Disciplines Required	None	Low	Med	High
Mechanical					**Materials Science**				
Electrical				**X**	**Computer Science**				**X**
Industrial					**Systems Eng**				**X**

Specialized Skills Required	Low	Med	High
Systems Engineering – Model and Architecture			**X**
Computer Science/Electrical Engineering – Artificial Intelligence/Machine Learning			**X**
Electrical Engineering/Software Engineering – Embedded Processor & Sensors			**X**
Engineering Management – Program Execution and Deliverables			**X**

U.S. Citizen Only **X** IP Agreement or NDA **X** Travel Required ☐